Introduction to Informatics and Algorithms

Chapter 1 | L1 Socle Commun Sciences de la Matière
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Student course notes | Two 90-minute lectures

Learning outcomes

- Distinguish data, information, hardware, software and a program.
- Identify inputs, processing and outputs in a scientific problem.
- Write a finite, ordered algorithm with variables and explicit outputs.
- Trace assignments and test a solution by hand.

Prerequisites

Basic arithmetic and the ability to read a simple scientific formula.

Session	Focus
1	Computer systems, programs and problem analysis
2	Algorithm structure, variables, operators and manual tracing

How to study: read the explanation, predict the result, trace the example, then attempt the checkpoint. Write your reasoning before consulting the teacher correction.

1.1 From measurements to information

Informatics concerns the representation and automatic processing of information. Data are recorded values or symbols. Information is the meaning obtained when those data are interpreted in context. A temperature value needs a unit and a measurement context. A computer can calculate a mean, but a scientist must judge whether the measurements and the model are appropriate.

Worked example

```
Measurements: 20, 22, 24 degrees C
Total: 20 + 22 + 24 = 66 degrees C
Mean: 66 / 3 = 22 degrees C
```

The three observations are the inputs. Addition and division are the processing. The mean and its unit are the output. A correct calculation does not prove that a sensor was correctly calibrated.

Check your understanding

What changes if the measurements are in kelvin?

Your explanation: _____

_____

1.2 How a computer processes a task

Hardware consists of physical components. Software consists of programs and associated instructions. Input devices provide data, the processor executes instructions, RAM holds active working data, and storage keeps files between sessions. Output devices communicate results. The operating system manages resources and provides services to applications.

Worked example

Input: keyboard or temperature sensor
Processor: executes the calculation
RAM: holds working values
Storage: saves the program and results
Output: screen or printer

RAM and persistent storage have different roles. Saving a file to storage is a deliberate action. Python is a programming language; IDLE is an editor and shell used to work with Python.

Check your understanding

Classify a keyboard, RAM, Windows and a Python script.

Your explanation: _____

_____

1.3 Algorithms and programs

An algorithm is a precise, ordered method that finishes for the inputs in its stated domain. A program expresses instructions in a language a computer can execute. The same algorithm can be expressed in natural language, pseudocode, a flowchart or Python. A formula describes a relationship; an algorithm also states what to read, calculate and report.

Worked example

```
Problem: calculate density
Given: mass in g and volume in mL, volume > 0
1. Read mass and volume.
2. Divide mass by volume.
3. Report density in g/mL.
```

State assumptions before calculating. At this stage, positive volume is a precondition. Chapter 3 will make the program check it automatically.

Check your understanding

Why is “calculate the answer” an inadequate algorithm?

Your explanation: _____

_____

1.4 A short history and a modern habit

The word algorithm derives from the Latinized name of al-Khwarizmi, a ninth-century scholar. Mechanical calculation, electronic computers, high-level languages and modern AI belong to a long development of computational methods. The central habit remains explicit reasoning: define the problem, represent data, specify steps and verify results. AI-generated code also requires testing.

Worked example

Problem analysis checklist
What is known?
What must be found?
Which formula or rule applies?
What values are allowed?
How can the result be checked?

Avoid confusing an algorithm with a logarithm. A logarithm is a mathematical function. An algorithm is a method, which may use a logarithm as one step.

Check your understanding

Is an AI-produced program necessarily correct? Explain.

Your explanation: _____

_____

1.5 The structure of pseudocode

This course uses ALGORITHM, VARIABLES, BEGIN and END to make the structure visible. READ obtains data and WRITE displays values or text. SET x TO expression means assignment. Quotation marks identify literal text; an unquoted variable name refers to its value. Pseudocode conventions vary, but each solution must use one consistent convention.

Worked example

```
ALGORITHM Density
VARIABLES mass, volume, density: REAL
BEGIN
  READ mass, volume
  SET density TO mass / volume
  WRITE "Density =", density, "g/mL"
END
```

The title identifies the task. Declarations explain the data. The instruction block performs the task. Preconditions: mass >= 0 and volume > 0. No Python execution is required in this chapter.

Check your understanding

What is the difference between WRITE "mass" and WRITE mass?

Your explanation: _____
_____

1.6 Variables, types and assignment

A variable names a value that can change during execution. A constant represents a value intended to remain unchanged. Integers represent counts, real values represent measurements, strings represent text, and Boolean values represent true or false. Assignment evaluates the right-hand expression using the current values, then associates the result with the name on the left.

Worked example

```
SET a TO 5
SET b TO a
SET a TO a + 2
WRITE a, b
```

After the first line a is 5. After the second, b is 5. Updating a does not retroactively change b. The output is 7, 5. For these numeric values, b keeps the value assigned to it.

Check your understanding

Trace SET x TO 3; SET y TO 2*x; SET x TO 10.

Your explanation: _____

_____

1.7 Expressions and logical questions

Arithmetic operators build calculations. Comparisons ask questions and produce Boolean results. AND requires both conditions to be true, OR requires at least one, and NOT reverses a Boolean result. Parentheses clarify the intended order. In pseudocode, use SET for assignment and = for equality. In Python, assignment uses = and equality uses ==.

Worked example

```
2 + 3 * 4 = 14
(2 + 3) * 4 = 20
For mass = 10 and volume = 2:
(mass >= 0) AND (volume > 0) is TRUE
```

For positive integers, DIV gives the integer quotient and MOD the remainder: 9 DIV 2 = 4 and 9 MOD 2 = 1. Python uses // and %. Negative-number details will be addressed in Chapter 2.

Check your understanding

For x = 0, evaluate (x > 0) OR (x = 0).

Your explanation: _____

_____

1.8 Tracing and testing before coding

A trace table records variable values after each instruction. Test a normal case, a meaningful boundary case and a case that violates a precondition. A test compares an expected result with an observed result. Do not merely record that an algorithm “works”. Use units and check whether the magnitude is plausible.

Worked example

Input: mass = 10 g, volume = 2 mL
Expected density: 5 g/mL
Boundary: mass = 0, volume = 2 gives 0
Outside domain: volume = 0
Decision: reject this input before dividing

The zero-volume case reveals a missing requirement; it must not be evaluated. In Chapter 1, record the precondition. Later, implement a guard.

Check your understanding

Write the steps for swapping x = 3 and y = 5.

Your explanation: _____
_____

Chapter synthesis and revision

- Can I distinguish data, information, hardware, software and a program?
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- Can I write a finite, ordered algorithm with variables and explicit outputs?
- Can I trace assignments and test a solution by hand?

English	French
input / output	entrée / sortie
assignment	affectation
condition / branch	condition / branche
loop / iteration	boucle / itération
trace / test	trace / test
pseudocode / flowchart	pseudo-code / organigramme

Reference reading

Python Software Foundation. Python 3 Tutorial: An Informal Introduction to Python. https://docs.python.org/3/tutorial/introduction.html

Python Software Foundation. More Control Flow Tools. https://docs.python.org/3/tutorial/controlflow.html

Python Software Foundation. math: Mathematical functions. https://docs.python.org/3/library/math.html

Python Software Foundation. IDLE: Python editor and shell. https://docs.python.org/3/library/idle.html